

IKT215 Assignment 1: Gaussian Naïve Bayes Classifier

All assignments are mandatory. Assignments equally contribute to 50% of your final grade.

Description

In the lectures, we learned about Bayes Classifier for binary classification, which can be formulated as:

$$\psi^*(\mathbf{x}) = \begin{cases} 1, & P(Y = 1)p(\mathbf{x} | Y = 1) > P(Y = 0)p(\mathbf{x} | Y = 0) \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

for all $\mathbf{x} \in R^d$

or we assign class 1, when

$$P(Y = 1)p(x_1 | Y = 1)p(x_2 | Y = 1) \dots p(x_d | Y = 1) > P(Y = 0)p(x_1 | Y = 0)p(x_2 | Y = 0) \dots p(x_d | Y = 0) \quad (2)$$

In multiclass classification the class is assigned using:

$$\psi^*(\mathbf{x}) = \arg \max_j P(Y = y_j | \mathbf{X} = \mathbf{x}) \quad (3)$$

By Bayes theorem,

$$P(Y = y_j | \mathbf{X} = \mathbf{x}) = \frac{P(\mathbf{X} = \mathbf{x} | Y = y_j)P(Y = y_j)}{P(\mathbf{X} = \mathbf{x})}$$

Or

$$P(Y = y_j | \mathbf{X} = \mathbf{x}) \propto P(Y = y_j)P(\mathbf{X} = \mathbf{x} | Y = y_j) \quad (4)$$

(4) in (3) implies, we assign the class y_i that maximizes

$$P(Y = y_j)p(x_1 | Y = y_j)p(x_2 | Y = y_j) \dots p(x_d | Y = y_j) \quad (5)$$

We can build a naïve classifier based on (5). The classifier is called 'naive' because it assumes that the features are conditionally independent given the class label. When we additionally assume that $P(\mathbf{X} = \mathbf{x} | Y = y_j)$ follows a Gaussian Normal Distribution, we get the Gaussian Naive Bayes variant of the classifier. Assuming a Gaussian Normal Distribution given by:

$$p(\mathbf{X} = \mathbf{x} | Y = y_j) = \frac{1}{\sigma_{i,j}\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x_i - \mu_{i,j}}{\sigma_{i,j}}\right)^2\right) \quad (6)$$

where $\mu_{i,j}$ and $\sigma_{i,j}$ are the mean and standard deviation, respectively, of the feature x_i given, they belong to class y_j .

Assignment Tasks:

1. Write your own Python code to implement a Gaussian Naïve Bayes Classifier from scratch on a synthetic binary classification dataset with 1000 samples and two features. **You may use NumPy/SciPy for mathematical operations and sklearn for dataset generation and train/test splitting, but do not use sklearn's GaussianNB classifier.**
 - Generate a synthetic dataset with two classes.
 - Split into train and test.
 - Implement the Gaussian Naïve Bayes algorithm.
 - Visualize: scatter plot of data with true class labels, decision boundary overlay, and confusion matrix
 - Report: accuracy, precision, and recall on the test set.
2. Extend your code to handle any number of features and classes (multi-class classification).
3. Test your code on the Iris flower dataset (<https://archive.ics.uci.edu/dataset/53/iris>), initially using the two sepal features and then using all four features. Visualize the results using appropriate plots (e.g. confusion matrix, decision boundary, etc.).
4. Discuss the procedures, results, assumptions, and obstacles you ran into while executing the above tasks and **prepare a professional report (3 pages maximum, excluding front matter if any)**. Include only the most relevant visualizations in the report.
5. Compile the report (PDF) and your code as a single zip file and name your file “lastname_firstname.zip” before uploading.

NB: Submit your code as a jupyter notebook file (“*.ipynb”) with outputs saved. Otherwise, scripts could run into failures due to dependencies on execution. Instructions for running a jupyter notebook are provided on Canvas.

Rubric

Task	Score (%)
Task 1	25
Task 2	20
Task 3	25
Task 4	20
Report Quality	10
Total	100

Use of Generative AI:

While the use of generative AI tools is not prohibited, it is strongly discouraged for this assignment. All reports must include a **dedicated disclaimer page** that honestly describes which AI tools (if any) were used and how they were applied, in accordance with [UiA's general guidelines](#). Remember that submitting AI-generated text without proper citation constitutes plagiarism and will be treated as such.

Late Submission Policy

Late submissions will incur a **5% penalty per calendar day**. Submissions will not be accepted more than five days after the deadline. Plan accordingly and contact the instructor in advance if you anticipate difficulties meeting the deadline.